

Exploratory Data Analysis Report

Titanic Dataset - Task 5

Objective: Extract patterns, trends, relationships, and anomalies using Python, Pandas, Matplotlib, and Seaborn.

Dataset Overview

Metric	Value
Rows	891
Original columns	12
Survived	342 (38.38%)
Did not survive	549 (61.62%)
Duplicate rows	0

- The dataset contains passenger demographics, passenger class, family information, ticket/fare details, cabin, embarkation port, and survival outcome.

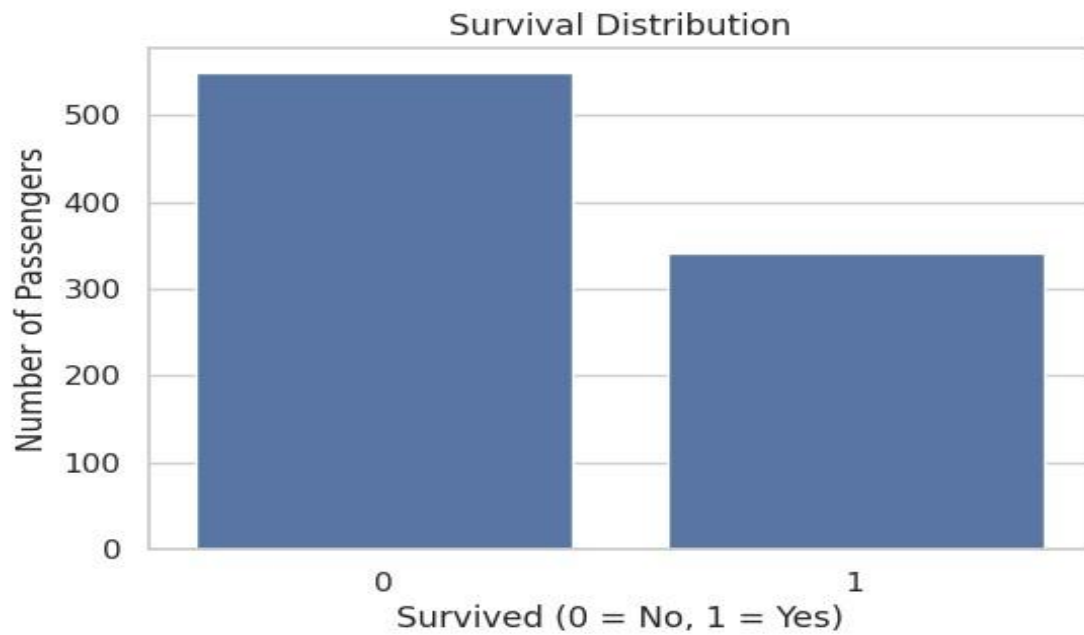
Data Quality and Preprocessing

- Before cleaning, missing values were: Age = 177, Cabin = 687, and Embarked = 2. No duplicate rows were present. Missing Age values were filled with the median and missing Embarked values with the mode. Cabin values were not guessed; instead, a CabinAvailable indicator was created to distinguish passengers with and without recorded cabin information.

Distribution Findings

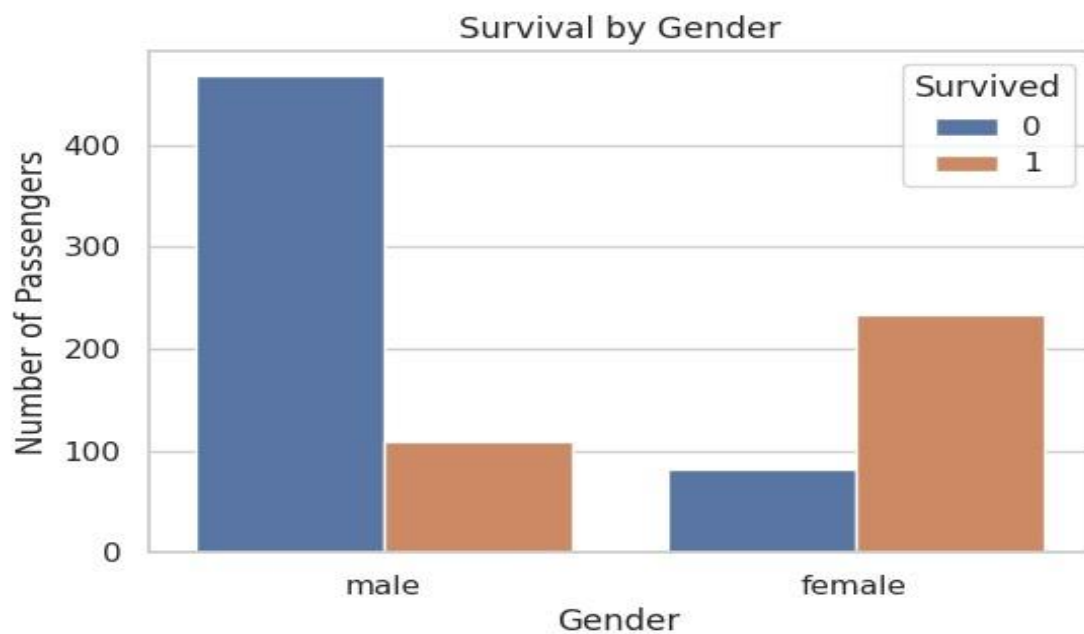
- The dataset contains 577 male and 314 female passengers.
- Third class is the largest group with 491 passengers, followed by first class (216) and second class (184).
- Southampton is the most common embarkation port with 644 passengers.
- Age is concentrated around young and middle adulthood.
- Fare is highly right-skewed and contains notable high-value outliers, with a maximum fare of 512.33.

Survival Distribution



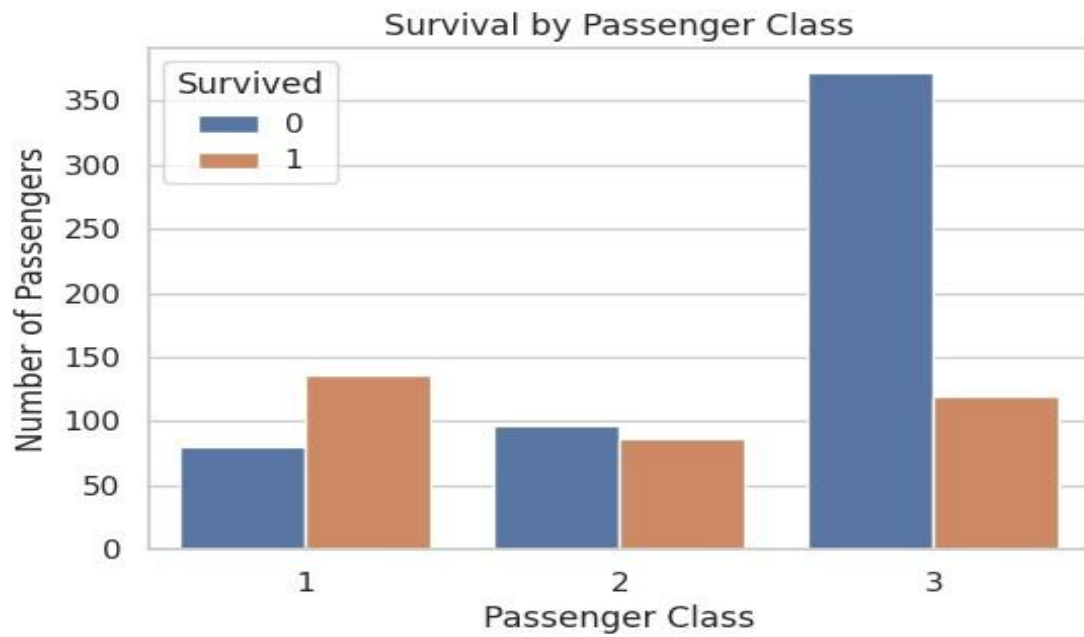
Observation: Non-survival is the majority outcome.

Survival by Gender



Observation: Female passengers show substantially higher survival than males.

Survival by Passenger Class



Observation: Survival varies strongly by passenger class.

Fare Distribution and Outliers



Observation: Several high-fare observations appear as outliers.

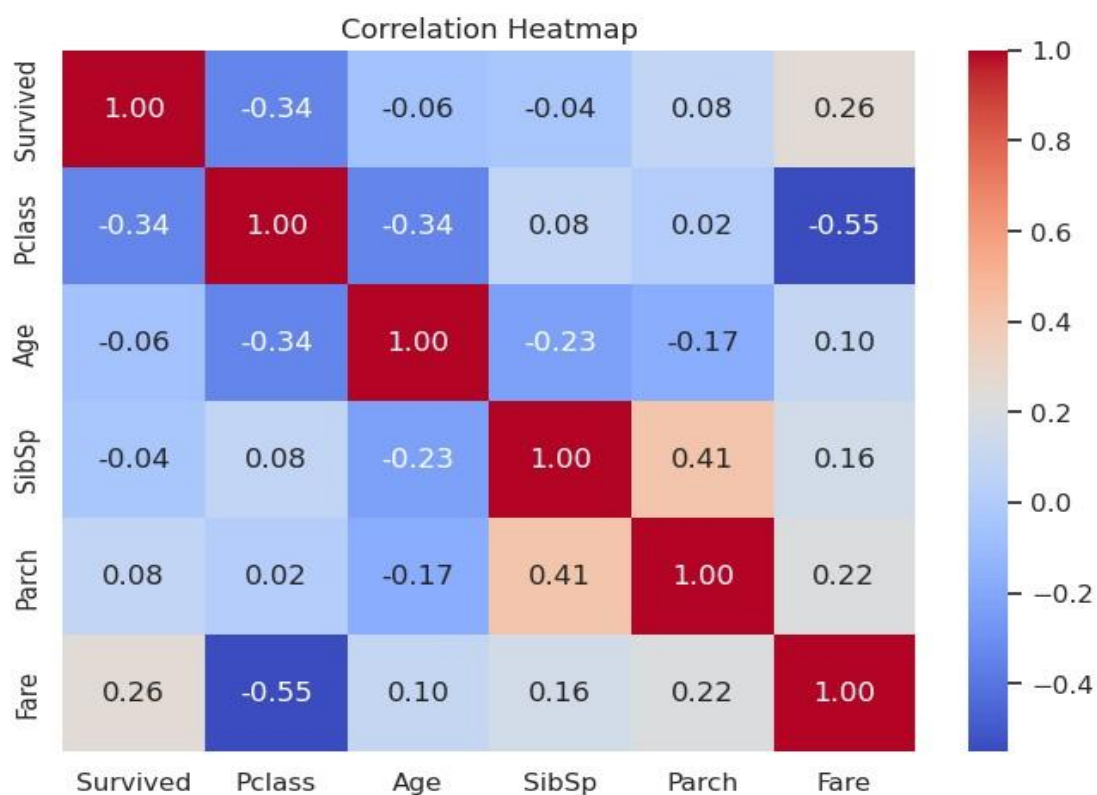
Relationship Analysis

Gender: Female survival was 74.20%, compared with 18.89% for males.

Gender and class: Female first-class survival was 96.81%, female second-class survival 92.11%, and female third-class survival 50.00%. Male first-, second-, and third-class survival rates were 36.89%, 15.74%, and 13.54% respectively.

Cabin availability: Passengers with recorded cabin information had a 66.67% survival rate versus 29.99% for those without recorded cabin information. This is an association and may reflect other factors such as passenger class.

Correlation Heatmap



Observation: Passenger class and fare show notable relationships with survival; correlations involving Age, SibSp, and Parch are comparatively weaker. Correlation does not establish causation.

Key Insights

- Only 38.38% of passengers survived, so non-survival was the majority outcome.
- Gender is one of the strongest visible factors: female survival was nearly four times the male survival rate.
- Passenger class is strongly associated with survival, with first-class passengers generally doing better than third-class passengers.

- The combination of gender and class is particularly informative: female first-class survival was about 96.81%, while male third-class survival was about 13.54%.
- Fare contains substantial outliers and is right-skewed, indicating a small number of unusually expensive tickets.
- Cabin missingness is very high (687/891), so treating missing cabin numbers as actual cabin values would be inappropriate.

Conclusion

The Titanic EDA demonstrates how descriptive statistics and visualizations can uncover meaningful patterns before predictive modeling. The clearest observed differences in survival are associated with gender and passenger class, while fare and cabin availability provide additional signals. The analysis also highlights the importance of handling missing data carefully and recognizing outliers rather than blindly removing or imputing them.

Tools used: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter/Google Colab.